

IEEE 30-Bus Battery Siting — Model Reference

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1 Use Case Overview

- **Network:** IEEE 30-bus, 41 branches, 6 generators (MATPOWER case30)
- **Batteries:** 4 units (30 MW / 120 MWh each), placement to be optimised
- **Search space:** $\binom{30}{4} = 27,405$ candidate placements
- **Objective:** minimise total 24-hour system dispatch cost
- **Horizon:** single 24-hour day, hourly resolution
- **Power flow:** DC approximation (PTDF-based, lossless, no reactive power)
- **Quantum siting:** 6 gen qubits + 30 bus qubits = **36 qubits** (Aer TN recommended)
- **Source:** <https://github.com/Square-Peg-Technologies/square-peg-quantum-challenge>

Key Files and Variables

File	Variable / Function	Description
ieee30.py	Case	Grid object: PTDF, fbar, power_demand
	Case.factors	24-hour demand multipliers (list of 24)
	CaseDescription.fbar	Branch flow limits (MW), 41 elements
	CaseDescription.branch	Branch r, x, b, ratio data
	CaseDescription.bus	Bus Pd, Qd base values
assets.py	GENERATORS	List of 6 generator dicts (bus, Pmin, Pmax, cost)
	BATTERIES	List of 4 battery dicts (power_mw, capacity_mwh, ...)
	DATACENTER_BUS	None (base case, no datacenter)
	DATACENTER_MW	0.0 (base case)
locations.py	GENERATOR_LOCATIONS	Dict {gen_index: bus}
	BATTERY_LOCATIONS	Placeholder; overridden by solver
case30.m	—	MATPOWER source data (retrieved June 2026)

2 Network: IEEE 30-Bus

2.1 Bus Data

Bus	Type	Base Pd (MW)	Base Qd (MVar)	Notes
1	Slack (3)	0.0	0.0	Generator only
2	PV (2)	21.7	12.7	Generator + load
3	PQ (1)	2.4	1.2	Load only
4	PQ (1)	7.6	1.6	Load only
5	PQ (1)	0.0	0.0	Shunt cap 0.19 pu
6	PQ (1)	0.0	0.0	Transit bus
7	PQ (1)	22.8	10.9	Load only
8	PQ (1)	30.0	30.0	Load only
9	PQ (1)	0.0	0.0	Transit bus
10	PQ (1)	5.8	2.0	Load only
11	PQ (1)	0.0	0.0	Transit bus
12	PQ (1)	11.2	7.5	Load only
13	PV (2)	0.0	0.0	Generator only
14	PQ (1)	6.2	1.6	Load only
15	PQ (1)	8.2	2.5	Load only
16	PQ (1)	3.5	1.8	Load only
17	PQ (1)	9.0	5.8	Load only
18	PQ (1)	3.2	0.9	Load only
19	PQ (1)	9.5	3.4	Load only
20	PQ (1)	2.2	0.7	Load only
21	PQ (1)	17.5	11.2	Load only
22	PV (2)	0.0	0.0	Generator only
23	PV (2)	3.2	1.6	Generator + load
24	PQ (1)	8.7	6.7	Shunt cap 0.04 pu
25	PQ (1)	0.0	0.0	Transit bus
26	PQ (1)	3.5	2.3	Load only
27	PV (2)	0.0	0.0	Generator only
28	PQ (1)	0.0	0.0	Transit bus
29	PQ (1)	2.4	0.9	Load only
30	PQ (1)	10.6	1.9	Load only
Total base load		189.2		

2.2 Branch Data

Br#	From-To	r (pu)	x (pu)	b (pu)	Limit (MW)	Ratio	Note
0	1-2	0.0200	0.0600	0.0300	130	—	Strong backbone
1	1-3	0.0500	0.1900	0.0200	130	—	
2	2-4	0.0600	0.1700	0.0200	65	—	
3	3-4	0.0100	0.0400	0.000	130	—	
4	2-5	0.0500	0.2000	0.0200	130	—	
5	2-6	0.0600	0.1800	0.0200	65	—	
6	4-6	0.0100	0.0400	0.000	90	—	
7	5-7	0.0500	0.1200	0.0100	70	—	
8	6-7	0.0300	0.0800	0.0100	130	—	Bottleneck to bus-8 cluster
9	6-8	0.0100	0.0400	0.000	32	—	
10	6-9	0.0000	0.2100	0.000	65	—	
11	6-10	0.0000	0.5600	0.000	32	—	
12	9-11	0.0000	0.2100	0.000	65	—	High-x path
13	9-10	0.0000	0.1100	0.000	65	—	
14	4-12	0.0000	0.2600	0.000	65	—	
15	12-13	0.0000	0.1400	0.000	65	—	
16	12-14	0.1200	0.2600	0.000	32	—	
17	12-15	0.0700	0.1300	0.000	32	—	
18	12-16	0.0900	0.2000	0.000	32	—	
19	14-15	0.2200	0.2000	0.000	16	—	
20	16-17	0.0800	0.1900	0.000	16	—	Weak peripheral link
21	15-18	0.1100	0.2200	0.000	16	—	
22	18-19	0.0600	0.1300	0.000	16	—	
23	19-20	0.0300	0.0700	0.000	32	—	
24	10-20	0.0900	0.2100	0.000	32	—	
25	10-17	0.0300	0.0800	0.000	32	—	
26	10-21	0.0300	0.0700	0.000	32	—	
27	10-22	0.0700	0.1500	0.000	32	—	
28	21-22	0.0100	0.0200	0.000	32	—	
29	15-23	0.1000	0.2000	0.000	16	—	
30	22-24	0.1200	0.1800	0.000	16	—	
31	23-24	0.1300	0.2700	0.000	16	—	
32	24-25	0.1900	0.3300	0.000	16	—	Peripheral feeder
33	25-26	0.2500	0.3800	0.000	16	—	
34	25-27	0.1100	0.2100	0.000	16	—	
35	28-27	0.0000	0.4000	0.000	65	—	
36	27-29	0.2200	0.4200	0.000	16	—	High-x tail
37	27-30	0.3200	0.6000	0.000	16	—	
38	29-30	0.2400	0.4500	0.000	16	—	
39	8-28	0.0600	0.2000	0.0200	32	—	
40	6-28	0.0200	0.0600	0.0100	32	—	

Key bottlenecks: branches 9 (6-8, 32 MW), 11 (6-10, 32 MW), 19/20/21 (16 MW peripheral feeders in the bus-14-20 cluster) bind at peak and produce non-zero shadow prices.

3 Generators

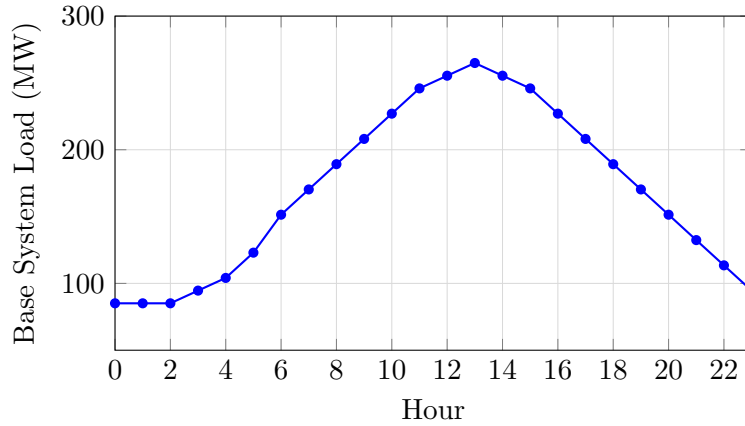
Six generators. Two cheap units (buses 1, 2), two medium units (buses 22, 27), two expensive units (buses 23, 13). Cost uses the MATPOWER case30 polynomial gencost ($a_g p^2 + b_g p + c_g$, with $c_g = 0$).

Name	Bus	Pmin (MW)	Pmax (MW)	a_g (\$/MW ² h)	b_g (\$/MWh)	Startup (\$)
Gen 1	1	10	80	0.02000	2.00	0
Gen 2	2	10	80	0.01750	1.75	0
Gen 3	22	10	50	0.06250	1.00	0
Gen 4	27	10	55	0.00834	3.25	0
Gen 5	23	5	30	0.02500	3.00	0
Gen 6	13	5	40	0.02500	3.00	0
Total installed		335 MW				

Note: Pmin raised from 0 (MATPOWER default) to small positive values so that unit-commitment decisions are meaningful. At peak demand (264.9 MW) the system has 26% reserve.

4 Load Profile

The 24-hour demand shape is defined in `ieee30.py` (`Case.factors`), a list of 24 hourly multipliers applied uniformly to all base bus loads. Source: synthetic daily shape scaled to the MATPOWER case30 base load of 189.2 MW. Peak system demand: $189.2 \times 1.40 = 264.9$ MW at Hour 13.



5 Battery Specifications

Four identical batteries: **30 MW / 120 MWh**, charge efficiency **90%** (discharge lossless), **initial SoC 120 MWh** (fully charged). Defined in `assets.py` as `BATTERIES`.

6 Problem Formulation

6.1 Objective

The cost function in `solvers/uc.py` supports a quadratic-plus-linear form:

$$\min \sum_g \sum_{t=1}^{24} [a_g \cdot p_{g,t}^2 + b_g \cdot p_{g,t} + c_g \cdot u_{g,t} + sc_g \cdot v_{g,t}]$$

In this use case all six generators carry the MATPOWER polynomial cost coefficients directly ($a_g > 0$). No-load costs c_g and startup costs sc_g are zero.

6.2 Decision Variables

- $p_{g,t} \geq 0$: generator output (MW)
- $u_{g,t} \in \{0, 1\}$: generator commitment (1 = online)
- $v_{g,t} \in \{0, 1\}$: startup indicator
- $r_{b,t}^+ \geq 0$: battery charge (MW)
- $r_{b,t}^- \geq 0$: battery discharge (MW)
- $\text{SoC}_{b,t} \geq 0$: state of charge (MWh)
- $z_{b,t} \in \{0, 1\}$: direction flag (1 = charging)

6.3 Constraints

Generator bounds:

$$P_g^{\min} \cdot u_{g,t} \leq p_{g,t} \leq P_g^{\max} \cdot u_{g,t}$$

Power balance (per hour):

$$\sum_g p_{g,t} + \sum_b (r_{b,t}^- - r_{b,t}^+) = \sum_i D_{i,t}$$

DC line flow limits (per branch ℓ , per hour):

$$-\bar{f}_\ell \leq \sum_i \text{PTDF}_{\ell,i} \cdot \text{net}_{i,t} \leq \bar{f}_\ell$$

Battery charge/discharge exclusion:

$$r_{b,t}^+ \leq P^{\text{bat}} \cdot z_{b,t}, \quad r_{b,t}^- \leq P^{\text{bat}} \cdot (1 - z_{b,t})$$

SoC bounds: $0 \leq \text{SoC}_{b,t} \leq 120$ MWh, with $\text{SoC}_{b,0} = 120$ MWh.

6.4 Siting Algorithm

The Benders decomposition (`siting_benders.py`) works as follows:

1. **Master problem** (SCIP MIP): selects which buses to place the 4 batteries on. Exactly one bus per battery; symmetry-breaking enforces ascending bus index.
2. **Subproblem** (`run_uc()`): given a fixed placement, solves the 24-hour UC/ED. Returns total cost Z .
3. **Cuts:** no-good cuts prevent re-visiting placements. L-shaped optimality cuts $\eta \geq Z \cdot (\sum_{(b,n) \in S^+} x_{b,n} - (M - 1))$ bound the master.
4. **Convergence:** master lower bound $\geq$ best known cost, or time limit.

An alternative joint MILP is in `siting_mip.py` (single SCIP solve, big-M linearisation of placement $\times$ dispatch).

7 Quantum Siting

The quantum siting solver (`solvers/quantum_siting.py`) encodes the joint (generator commitment, battery placement) decision as a **36-qubit** QAOA/VQA problem.

- **Qubit layout:** $u_0 \dots u_5$ (6 generator bits) $\parallel s_0 \dots s_{29}$ (30 bus bits)
- **Circuit:** butterfly ansatz, L layers, $2Ln$ parameters
- **Optimiser:** COBYLA with plateau detection (patience = $\max(50, 2Ln)$ evals)
- **Recommended backend:** `aer_tn` (Aer tensor-network MPS simulator); statevector simulation of 36 qubits requires ~ 1 TB RAM and is not practical

Proxy cost function $Q(u, s)$ encodes:

1. $c_{\min}(u)$: minimum-cost dispatch at $p_g = P_g^{\min}$ for committed generators
2. $\lambda_1 \cdot (\sum_i s_i - B)^2$: battery-budget penalty (exactly B batteries)
3. $\lambda_2 \cdot \max(0, D - \sum_g P_g^{\max} u_g)^2$: generator shortfall penalty
4. $-P_{\text{loc}}(s)$: congestion relief bonus from PTDF $\times$ shadow-price signal

8 Comparison

Method	Solver	Qubit / binary vars	Typical runtime
ED (fixed placement)	CVXPY / Clarabel	none	<1 s
UC (fixed placement)	CVXPY / SCIP	$6 \times 24 = 144$ binary	5–30 s
Siting (Benders)	SCIP + UC	none (iterative)	60–300 s
Siting (MIP)	SCIP	$30 \times 4 = 120$ binary	60–600 s
Quantum (D-Wave SA)	dimod SA	36 qubits (BQM)	5–20 s
Quantum (Qiskit SV)	Aer statevector	36 qubits	not practical (~ 1 TB)
Quantum (Aer TN)	Aer MPS	36 qubits	60–300 s